May 11, 2026

Dr. Peter H. Thrall
Editor-in-Chief, *Ecology Letters*
CSIRO National Collections & Marine Infrastructure
GPO Box 1600, Canberra ACT 2601, Australia

Dear Dr. Thrall,

We are pleased to submit our manuscript, *"Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses"*, for consideration as a Letter in *Ecology Letters*.

Understanding when and why mutualisms break down under environmental stress is a central challenge in ecology, with increasing urgency as climate change pushes species into novel conditions. Despite extensive theory on the stress dependence of symbiotic benefits, rigorous empirical tests remain rare, particularly those that move beyond single host species or controlled settings to evaluate symbiont effects across the full range of conditions hosts naturally encounter. Our study addresses this gap through an integrative design. We conducted common garden experiments with three grass species hosting vertically transmitted fungal endophytes (*Epichlöe* spp.) across a natural aridity gradient spanning seven sites in the south-central United States, with herbivore exclusion treatments nested within sites. We analyzed demographic vital rates using hierarchical models that partition variation across species, sites, and treatments, providing resolution unattainable in simpler designs. A complementary greenhouse experiment, which independently manipulated soil origin, allowed us to isolate mechanisms underlying field patterns. By combining multi-species field experiments, demographic modelling, and controlled greenhouse tests, our study provides a rare, mechanistic evaluation of context dependent symbiosis. We believe this approach allows us to address both generality and underlying drivers, questions of broad relevance to the readership of *Ecology Letters*.

We have opted out of double-blind review because this paper builds upon our previous work, so our identities could be easily deduced. Our identities are also evident from our GitHub repository, where we direct reviewers to find all data and code necessary to reproduce our analyses.

Sincerely yours,

Jacob K. Moutouama
On behalf of all co-authors